CG (Conjugate Gradient)

linear CG e.g. $\min f = \frac{1}{2}x^TAx - \langle x, b\rangle$

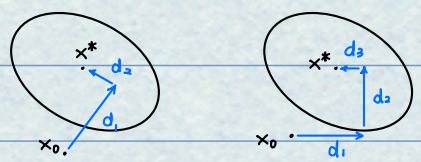
下降 $x_{k+1} = x_k + \alpha_k d_k$ $\quad \alpha_k = -\frac{\langle r_k, p_k\rangle}{\langle p_k, p_k\rangle_A}$

确定方向 $d_{k+1} = -r_k + \beta_k d_k$ $\quad \beta_k = \frac{\langle r_k, d_k\rangle_A}{\langle d_k, d_k\rangle_A}$

Def. $d_1 \cdots d_N$ 互为 conjugate

$\langle d_i, d_j\rangle_A = d_i^T A d_j = 0$ 加权正交

Thm. conjugate $\{d_n\}$ 用 N 步收敛到 x^* 沿着椭圆的基



CG, 2步 $\quad$ 坐标梯度

Nonlinear CG e.g. FR (Fletcher-Reeves)

下降 $x_{k+1} = x_k + \alpha_k d_k$

确定方向 $d_{k+1} = -\nabla f_k + \beta_k d_k$ $\quad \beta_k^{FR} = \frac{\|\nabla f_{k+1}\|^2}{\|\nabla f_k\|^2}$